

Diagram gallery

This document exercises every documented feature of Lout's @Diag package (the diag library, from chapter 9 of the Lout User's Guide). Each example uses the 'lout raw passthrough fence – mdlout does not yet have a markdown shortcut for diagrams.

mdlout auto-detects @Diag / @Tree / @Node / @Link / shape macros in raw Lout fenced blocks and adds @SysInclude { diag } to the preamble. Inside @Diag { ... } the available shape macros are @Box (the default for @Node), @CurveBox, @ShadowBox, @Square, @Diamond, @Polygon, @Isosceles, @Ellipse, @Circle, plus @Tree for tree diagrams.

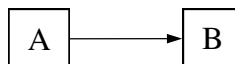
Arrowstyles

Lout's diagram links support ten arrowhead styles. Each example below draws two nodes connected by a directed link, varying only the arrowstyle option.

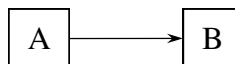
solid (filled triangle, the default)



solidwithbar (filled triangle behind a perpendicular bar)



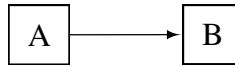
halfopen (open V, only one barb)



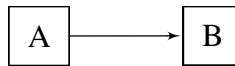
open (open V, both barbs)



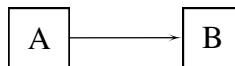
curvedsolid



curvedhalfopen



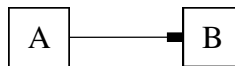
curvedopen



circle



box



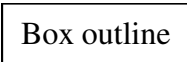
many



Node outlines

@Diag offers several built-in outlines via shape macros. Each one is shorthand for @Node outline { name }.

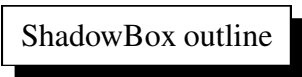
Box (@Box / @Node, the default)

A rectangular box with a thin black border and the text "Box outline" centered inside.

CurveBox (rounded)

A rounded rectangular box with a thin black border and the text "CurveBox outline" centered inside.

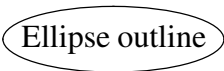
ShadowBox

A rectangular box with a thin black border and the text "ShadowBox outline" centered inside. The box has a thick black shadow cast to its right and bottom.

Square

A square box with a thin black border and the text "Sq" centered inside.

Ellipse

An elliptical box with a thin black border and the text "Ellipse outline" centered inside.

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Circle



Cir

Diamond



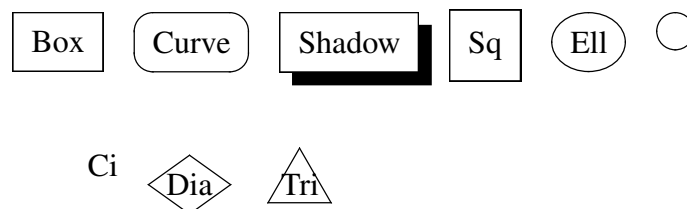
Polygon (3-sided -> triangle)



Isosceles (a tall triangle)



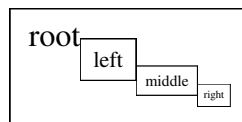
Row of all available shapes



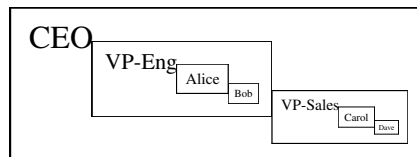
Tree diagrams (@Tree inside @Diag)

@Tree lays out a node with subtree children attached underneath via @LeftSub, @RightSub, @Sub. Inside it, the same shape macros work as @TNode-flavoured nodes.

Simple two-level tree



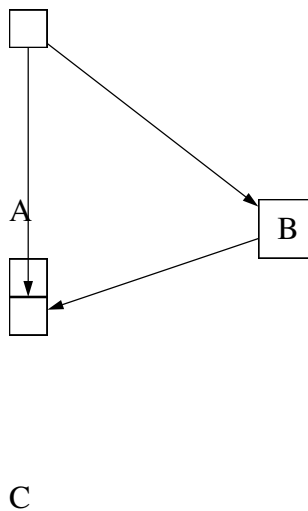
Three-level tree



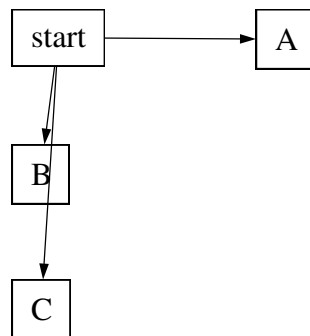
Multi-link diagrams

Three or more nodes connected by various paths.

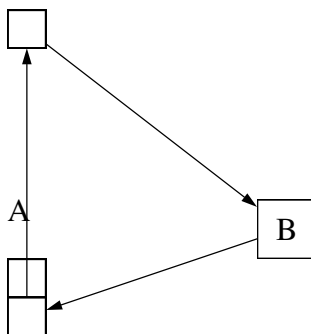
Triangle of three nodes, all pairs linked



Branching graph (1-to-many)



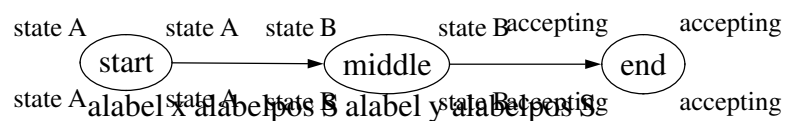
Cycle with labelled links



C
 alabel a-to-b alabelpos N alabel b-to-c alabelpos E alabel c-to-a alabelpos W

@Ellipse with tag labels (state machine)

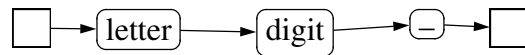
State-machine-style example: ellipse nodes with external labels at their boundary.



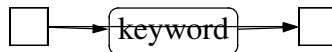
Syntax-diagram style (railroad)

Lout's diag library does not ship a dedicated `@SyntaxDiag` macro, but the same effect is straightforward to build with `@CurveBox` for terminal boxes and explicit `@Link` paths. The classic page-220 examples become:

Identifier sequence: letter -> digit -> underscore



Optional element (skip path)



End

This concludes the `@Diag` gallery. If your build succeeded, every section above produced a rendered diagram in the output HTML/PDF.